

MCQS

Sta630 midterm preparation file

ZB

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

In the name of God , Most
Gracious, Most merciful



STA630

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1. Which of the following is the first step in the research process?

- a) Data collection
- b) Problem identification
- c) Literature review
- d) Hypothesis formulation

Answer: b) Problem identification

2. A research hypothesis is:

- a) A proven fact
- b) A testable statement
- c) A statistical analysis
- d) An untestable assumption

Answer: b) A testable statement

3. Which type of research aims to solve a specific problem?

- a) Basic research
- b) Applied research
- c) Exploratory research
- d) Descriptive research

Answer: b) Applied research

4. In qualitative research, the primary focus is on:

- a) Statistical analysis
- b) Numerical data
- c) Subjective interpretation
- d) Hypothesis testing

Answer: c) Subjective interpretation

5. A variable that the researcher manipulates is called:

- a) Dependent variable
- b) Independent variable
- c) Confounding variable
- d) Extraneous variable

Answer: b) Independent variable

6. A literature review helps in:

- a) Formulating hypotheses
- b) Identifying gaps in research
- c) Establishing theoretical frameworks
- d) All of the above

Answer: d) All of the above

7. Which of the following sampling methods is non-probability based?

- a) Simple random sampling
- b) Stratified sampling
- c) Convenience sampling
- d) Cluster sampling

Answer: c) Convenience sampling

8. The degree to which a research instrument measures what it intends to measure is called:

- a) Reliability
- b) Validity
- c) Accuracy
- d) Consistency

Answer: b) Validity

9. When two variables move in opposite directions, this is referred to as:

- a) Positive correlation
- b) Negative correlation
- c) No correlation
- d) Causal relationship

Answer: b) Negative correlation

10. A structured questionnaire primarily includes:

- a) Open-ended questions
- b) Closed-ended questions
- c) Observational notes
- d) Interviews

Answer: b) Closed-ended questions

11. An operational definition:

- a) Provides the dictionary meaning of a concept
- b) Explains how a concept will be measured
- c) Describes the population
- d) Summarizes the research problem

Answer: b) Explains how a concept will be measured

12. The group to which the results of a study apply is called:

- a) Sample
- b) Population
- c) Subgroup
- d) Control group

Answer: b) Population

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13. In experimental research, the group that does not receive treatment is called:

- a) Experimental group
- b) Control group
- c) Random group
- d) Placebo group

Answer: b) Control group

14. Which of the following is a primary source of data?

- a) Journal articles
- b) Surveys
- c) Textbooks
- d) Literature reviews

Answer: b) Surveys

15. The measure of central tendency that represents the middle value is:

- a) Mean
- b) Median
- c) Mode
- d) Range

Answer: b) Median

16. Which of the following is a qualitative research method?

- a) Experiment
- b) Case study
- c) Survey
- d) Hypothesis testing

Answer: b) Case study

17. A hypothesis stating there is no relationship between variables is called:

- a) Alternative hypothesis
- b) Null hypothesis
- c) Research hypothesis
- d) Statistical hypothesis

Answer: b) Null hypothesis

18. What is the purpose of a pilot study?

- a) To collect large-scale data
- b) To test the feasibility of the study
- c) To develop a hypothesis
- d) To analyze results

Answer: b) To test the feasibility of the study

19. In descriptive research, the primary objective is to:

- a) Establish cause and effect
- b) Explore relationships
- c) Describe phenomena
- d) Test hypotheses

Answer: c) Describe phenomena

20. Which term refers to consistency in measurement?

- a) Validity
- b) Reliability
- c) Accuracy
- d) Generalizability

Answer: b) Reliability

21. A type of research that involves the collection of numerical data is called:

- a) Qualitative research
- b) Quantitative research
- c) Exploratory research
- d) Case study

Answer: b) Quantitative research

22. A double-blind study ensures:

- a) Researchers know the treatment group
- b) Both participants and researchers are unaware of group assignments
- c) Only participants know the treatment
- d) A second researcher verifies results

Answer: b) Both participants and researchers are unaware of group assignments

23. The most appropriate graphical representation of categorical data is:

- a) Histogram
- b) Pie chart
- c) Scatter plot
- d) Line graph

Answer: b) Pie chart

24. Which type of research involves the study of past records?

- a) Experimental research
- b) Historical research
- c) Correlational research
- d) Longitudinal research

Answer: b) Historical research

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25. The process of assigning numbers to variables is called:

- a) Classification
- b) Measurement
- c) Quantification
- d) Validation

Answer: b) Measurement

26. Which of the following is an advantage of secondary data?

- a) More accurate than primary data
- b) Less expensive and time-consuming
- c) Specifically tailored to the research problem
- d) Easier to validate

Answer: b) Less expensive and time-consuming

27. A systematic random sample requires:

- a) A fixed interval for selection
- b) Stratification of the population
- c) Convenience in sampling
- d) A non-representative population

Answer: a) A fixed interval for selection

28. An extraneous variable is:

- a) The variable being manipulated
- b) A variable not controlled but influencing results
- c) A variable irrelevant to the study
- d) The dependent variable

Answer: b) A variable not controlled but influencing results

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29. The main goal of inferential statistics is to:

- a) Describe data
- b) Predict relationships
- c) Generalize findings to a population
- d) Summarize numerical data

Answer: c) Generalize findings to a population

30. When respondents provide socially desirable answers, it leads to:

- a) Sampling bias
- b) Non-response error
- c) Social desirability bias
- d) Measurement error

Answer: c) Social desirability bias

31. What is the term for a small-scale version of a study?

- a) Pilot study
- b) Pre-study
- c) Mock study
- d) Feasibility study

Answer: a) Pilot study

32. Which of the following is an example of a qualitative data collection method?

- a) Questionnaire
- b) Interview
- c) Experiment
- d) Survey

Answer: b) Interview

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33. Which sampling method divides the population into subgroups and selects samples from each?

- a) Cluster sampling**
- b) Stratified sampling**
- c) Simple random sampling**
- d) Systematic sampling**

Answer: b) Stratified sampling

34. What is the purpose of the research design?

- a) To define the population**
- b) To outline the research methods**
- c) To analyze data**
- d) To summarize results**

Answer: b) To outline the research methods

35. The term "population" in research refers to:

- a) The group from which a sample is drawn**
- b) A large sample size**
- c) The experimental group only**
- d) All individuals with shared characteristics relevant to the study**

Answer: d) All individuals with shared characteristics relevant to the study

36. The Hawthorne effect refers to:

- a) Errors in data collection**
- b) Participants changing behavior due to awareness of being observed**
- c) Bias introduced by researchers**
- d) Statistical errors in analysis**

Answer: b) Participants changing behavior due to awareness of being observed

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37. The dependent variable in a study is:

- a) Manipulated by the researcher
- b) Measured as the outcome of interest
- c) Always numerical
- d) Not influenced by other variables

Answer: b) Measured as the outcome of interest

38. Which of the following is not a scale of measurement?

- a) Nominal
- b) Ordinal
- c) Ratio
- d) Quantitative

Answer: d) Quantitative

39. Which type of validity examines the extent to which the findings can be generalized?

- a) Construct validity
- b) External validity
- c) Internal validity
- d) Content validity

Answer: b) External validity

40. A hypothesis that predicts the direction of the relationship is called:

- a) Null hypothesis
- b) Non-directional hypothesis
- c) Directional hypothesis
- d) Complex hypothesis

Answer: c) Directional hypothesis

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41. What is a major drawback of longitudinal studies?

- a) Lack of reliability
- b) High cost and time consumption
- c) Poor generalizability
- d) Limited access to participants

Answer: b) High cost and time consumption

42. An essential characteristic of a good research problem is:

- a) Ambiguity
- b) Specificity
- c) Complexity
- d) Generalization

Answer: b) Specificity

43. In correlational research, the goal is to:

- a) Determine causation
- b) Establish relationships between variables
- c) Test hypotheses
- d) Conduct experiments

Answer: b) Establish relationships between variables

44. What is triangulation in research?

- a) Using multiple research methods or data sources
- b) Limiting bias in sampling
- c) Measuring three variables
- d) Analyzing data using three techniques

Answer: a) Using multiple research methods or data sources

45. Which of the following is an example of ordinal data?

- a) Blood type
- b) Height in centimeters
- c) Education level (e.g., high school, college, postgraduate)
- d) Temperature in Celsius

Answer: c) Education level (e.g., high school, college, postgraduate)

46. A census is:

- a) Sampling a subset of the population
- b) Collecting data from every member of the population
- c) Analyzing historical data
- d) A qualitative method

Answer: b) Collecting data from every member of the population

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WhatsApp:

<https://whatsapp.com/channel/0029VaODheTDp2Q8YVco0N2p>

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Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.

Therefore, for passing a course/subject, student shall fulfill the following:

- Secure minimum **20%** score in Formative Assessments/Mid terms
- Secure minimum **20%** score in Final Term Examinations.
- Secure at least **40%** marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?

A: Follow these steps:

1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).
2. Each subject typically has 3 credit hours.
3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.

Examples:

21 credit hours = 7 subjects
18 credit hours = 6 subjects

15 credit hours = 5 subjects

Remember, the university may allow different credit hours for each student, so check your specific allowance.

By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

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Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed

Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish
krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat
Fame

Moat

**Humy pta hona chaheay Moat tab ani ha jab Allah
chahy us se pehly puri dunya bi ak taraf ho ke apko
marna chahy to apka kuch nahe beggar saqti**

Risk/Dolat

**Risk Dolat sb Allah pak ke hath m ha puri dunya ak
taraf b ho jay na apsy apka risk cheen saqti ha na de
saqti ha Ye sirf ALLAH pak ky hath m ha**

Izat Zilat shohrat Fame

**Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b
apko zra brabr b damage nahe kr saqti Agr ALLAH**

**Apko izat dena chahay to puri dunya ak taraf ho ke
b apko 1% b nuksan nahe pohancha saqti**

Phr Dar ks bat ka ??????????????????

**Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne
mat jhuko puri dunya mel k b apka kuch b nahe
beggar saqti**

ZB
MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family

Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my campaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha

Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us

m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us

Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

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ZB

Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

..... **MY Family**

**ust Relax and focus! Exams are not difficult. Put in your
100% effort and trust in Allah.**

..... **AL-Quran**

**"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"**

May Allah bless you